

Arman Sakif Chowdhury

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Technical Skills

Programming Languages: Python, SQL, R, Java, JavaScript, C++, C, Dart, Bash.

Data Engineering: ETL/ELT Pipelines, Data Modeling, Change Data Capture (CDC), Data Acquisition, Data Cleaning, PySpark, Apache Spark, Databricks, Microsoft Fabric, SQL Server, Snowflake, Azure Data Lake, SAP Datasphere.

Machine Learning & Data Science: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, NLP, Data Augmentation.

AI & Agentic Systems: MCP Development, LLM Applications, CopilotKit, AG-UI, A2UI, AI Solution Development.

Business Intelligence & Visualization: Power BI, DAX, Power Query, Tableau, Microsoft Excel, Matplotlib, Seaborn, KPI Reporting, Data Storytelling.

Web & Application Development: HTML, CSS, React, Node.js, Django, Flask, Flutter, MongoDB, Selenium, BeautifulSoup, ANTLR.

Cloud, Tools & Practices: Azure, AWS (EC2, ECS, RDS), Kubernetes, Git, CI/CD, Unit Testing, Jupyter Notebook, Google Colab, Postman, Agile/Scrum.

Education

Master of Applied Computing (AI Stream)

Sep 2024 – Apr 2026

University of Windsor, Windsor, ON

- Cumulative average: 86.11% — all degree requirements completed (33 credit hours).
- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Advanced Database Topics, Distributed Systems, Advanced Computing Concepts, Advanced Software Engineering.

Bachelor of Science in Computer Science and Engineering

Jan 2019 – Jun 2023

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

- CGPA: 3.44 / 4.00 (medium of instruction: English).
- Relevant Coursework: Artificial Intelligence, Pattern Recognition, Soft Computing, Digital Image Processing, Distributed Database Systems, Database Systems, Algorithms, Data Structures, Formal Languages & Compilers, Mathematical Analysis for Computer Science.

Work Experience

AI Solutions and BI Developer Co-op

Sep 2025 – Present

FGF Brands, North York, ON

Technologies: Power BI, Databricks, PySpark, SQL Server, Azure, Python, MCP, AI Solutions

- Designed and published interactive Power BI dashboards across Dev, UAT, and Production environments, while redesigning deployment pipelines to reduce deployment time and manual effort by 60%.
- Built an automated data sync pipeline in PySpark on Databricks to replicate new records from a Databricks view into an on-premise SQL Server table, engineering custom change-data-capture logic that diffs source and destination datasets to append only deltas where no native CDC existed.
- Accelerated pipeline logging from 5+ minutes to near-instant by converting PySpark DataFrames to Pandas after the initial read, eliminating repeated re-materialization of a multi-join upstream view on every logging operation.
- Developed a custom Power BI MCP server on Power BI REST APIs, enabling natural-language querying of enterprise semantic models while enforcing security and answer accuracy through workspace scoping, table-and-column allowlists, and read-only access controls.

- Collaborated across three concurrent project teams in distinct roles: AI adoption researcher on the core AI Solutions team, Power BI developer in the SMI agile pod, and data engineer building master data in SAP Datasphere for a dynamic pricing initiative.

Python Developer Consultant

Sep 2023 – Aug 2024

Techscope, New York, USA (Remote)

Technologies: Python, Pandas, SQL, JSON, ANTLR

- Automated the conversion of unstructured client input into structured JSON by authoring custom ANTLR grammars and Python parse-tree visitors, replacing a manual transformation workflow and enabling downstream pipeline ingestion.
- Devised SQL-based data ETL (Extract, Transform, Load) pipelines to optimize data processing workflows on a transactional database of over 100,000 entries.
- Performed comprehensive unit testing using Python's unittest framework, reducing error rates by 25%, enabling early bug detection, and enhancing software reliability.
- Engaged with clients directly to gather requirements, analyze business processes, and integrated them into team sprints to facilitate collaboration.

Back-End Developer & Customer Support

May 2020 – Aug 2023

ProyogLab, Dhaka, Bangladesh

Technologies: React, Node.js, Python, Power BI, and Data Warehouse

- Provided hands-on technical support to customers, assisting with Arduino-based hardware projects and troubleshooting embedded systems issues.
- Managed day-to-day sales operations in a startup retail environment, helping scale product distribution from hobbyist kits to educational bundles.
- Played a key role in transitioning company to an online model, contributing to website architecture and backend functionality, including visual dashboard using modern web technologies.
- Developed and maintained backend systems for e-commerce operations, including inventory management, order processing, and data integration. Generated monthly reports using Power BI tools to report progress.

Publications

Tackling Fake News in Bengali: Unraveling the Impact of Summarization vs. Augmentation on Pre-trained Language Models

2023

Chowdhury, A. S., Shahariar, G. M., Aziz, A. T., Alam, S. M., Sheikh, M. A., & Belal, T. A. (2023)
arXiv preprint arXiv:2307.06979.

Projects

Predicting Movie Ratings from Plot Summaries [\[GitHub\]](#)

Feb 2025 - Mar 2025

Team Research, University of Windsor, ON

Technologies: Python, Pandas, Pytorch, Scikit-learn, Selenium and BeautifulSoup

- Conceptualized a Large Language Model (BERT) to predict movie success using pre-release data, specifically only using plot summaries as input.
- Utilized two publicly available corpora and created a custom dataset with 17,877 records by scraping the IMDB website, ensuring a comprehensive dataset for analysis.
- Exceeded baseline model performance by 11% with our 81% F1-score and deployed the optimized model on Huggingface Spaces for seamless accessibility and evaluation.

Mitigating Bias in Natural Language Understanding using GNN [\[GitHub\]](#)

Sep 2024 - Dec 2024

Team Research, University of Windsor, ON

Technologies: Python, Pandas, Pytorch, Scikit-learn, and Kagglehub

- Formulated a Graph Neural Networks (GNN) model to mitigate racial bias in natural language understanding by either removing less correlated sensitive attributes or assigning lower weights to them.
- Demonstrated the feasibility of building a computationally expensive model on low-resource platforms by utilizing only a CPU on Google Colaboratory.
- Maintained competitive performance by achieving 87.47% accuracy, closely matching state-of-the-art models (91%) that do not employ data anonymization.

Bengali Fake News Detection [\[Journal\]](#)

Aug 2022 - Sep 2025

Team Research, Ahsanullah University of Science and Technology, Bangladesh

Technologies: Python, PyTorch, Hugging Face Transformers, BERT, NLP, Data Augmentation

- Led a four-member research team through peer review to a first-author journal publication, classifying fake news in low-resource Bengali using five pre-trained transformer language models.
- Fine-tuned BanglaBERT and mBERT with summarization and data augmentation pipelines, achieving 96–97% accuracy on two benchmark test sets and 86% on a third set reserved for generalization evaluation.
- Curated an expanded training corpus by translating English news articles to offset Bengali fake-news scarcity, and summarized long articles to overcome BERT's token-length limitations.

Churn BigML: Telecom Customer Churn Prediction [\[GitHub\]](#)

Feb 2023 - May 2023

Team Research, University of Windsor, ON

Technologies: Python, Scikit-learn, LightGBM, XGBoost, Pandas

- Benchmarked eight classification algorithms (KNN, SVM, Random Forest, AdaBoost, Gradient Boosting, LGBM, XGBoost) on a 3,333-instance telecom dataset, reaching 95.74% accuracy with LGBM and XGBoost.
- Identified the strongest churn drivers through feature importance and correlation analysis, translating model results into actionable customer retention insights.

Related Technical Training

Agile Foundations – The Rise of Knowledge Workers (LinkedIn Learning)

Jan 2025

- Acquired hands-on understanding of Agile principles, Scrum roles, and iterative development, emphasizing team collaboration, adaptability, and continuous improvement in dynamic, knowledge-driven environments.

Flutter and Dart – The Complete Guide

Oct 2024

Academind by Maximilian Schwarzmüller (Udemy)

Ultimate AWS Certified Solutions Architect (Udemy)

Jun 2024

- Gained experience in designing and deploying scalable, secure, and cost-effective AWS cloud solutions and became familiar with services like EC2, S3, Lambda, RDS and IAM.

Spark and Python for Big Data with PySpark (Udemy)

Oct 2023

- Enhanced proficiency in distributed computing, data wrangling, optimization techniques for big data workflows, and overall knowledge on processing and analyzing large-scale datasets

100 Days of Code: The Complete Python Pro Bootcamp (Udemy)

Sep 2023

- Strengthened Python programming skills through hands-on projects covering automation, web development, and data science while using libraries like Flask, Pandas, Selenium, and TensorFlow for real-world applications.

Extracurricular Activities

VentureU Bootcamp

Feb 18-22, 2025

Office of Innovation, Partnerships and Entrepreneurship, University of Windsor, ON

- Benefited from expert-led 5-day-long bootcamp, enhancing problem-solving capabilities and fostering the development of practical, applicable solutions.
- Participated in team-based problem-solving and demonstrated presentation skills through competitive pitch competitions, demonstrating the ability to effectively communicate complex ideas.

Event Volunteer

Oct 2024 - Present

University Community Church, Windsor, ON

- Assisted with setup, serving, dishwashing and cleanup of Sunday meals for students.

International Students Mentor

Dec 2024 - Jan 2025

International Student Centre, University of Windsor, ON

- Mentored 15 international students at the University of Windsor by answering questions and assisting with transition.